# SAKSH MENON

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## EDUCATION

**Bachelor of Science, Computer Science** Expected Graduation: May 2026
*University of Cincinnati, Cincinnati, OH – 3.8 GPA*
**Honors:** National Society of Leadership and Success Honor Society, CEAS International Outreach Scholarship, and UC Global Scholarship
**Relevant Coursework:** Deep Learning, Applied AI & Machine Learning, Data Structures, Discrete Mathematics, Statistics & Probability, Linear Algebra

## PROFESSIONAL EXPERIENCE

**Machine Learning Intern – Bioinformatics** August 2024 – December 2024
*Cincinnati Children's Hospital, Cincinnati, OH*

- Designed a user-friendly, efficient tool for biologists to discover and study similarities between two or more organisms by leveraging **2000-4000+ ProtT5** PyTorch embeddings, vector cosine similarities, and four additional search algorithms
- Reduced similarity search times by **40%** compared to industry standards, achieved with an **Azure Kubernetes Services** workflow, which involved creating a new metric to increase confidence in results after preprocessing and transformations on distributed systems
- Explored several machine learning models, including **Protein Language Models, SVC, XGBoost, Ridge**, and **Lasso** to generate predictions on fatal and non-fatal amino acid mutations

**Research Intern – Fault Hunter Software Development** January 2024 – April 2024
*University of Cincinnati, Cincinnati, OH*

- Pioneered a pattern recognition technique deployed on Linux servers with port forwarding to enhance efficiency and effectiveness of vulnerability detection in embedded systems, specifically by addressing highly skewed data with a label ratio of 50:1
- Enhanced and tested an application to identify and mitigate over seven vulnerability patterns in both **C/C++** and **Assembly** code, assisted in expanding software capabilities, allowing us to exceed benchmarks
- Incorporated constructive criticism from team meetings, code reviews, and mentorship to modify code and experiment with new approaches, which helped foster a culture of continuous improvement and innovation

**Student Co-op – Software Development** January 2023 – April 2023
*Cincinnati Children's Hospital, Cincinnati, OH*

- Extracted insights from over **500,000** readings of fragile X syndrome patient eye tracking data, accomplished with an optimized custom data analysis pipeline, which contributing to the success of multiple research projects for a **billion-dollar** firm
- Enhanced productivity by reducing computation times by over **50%** with structured, low overhead, and reusable parallel computing techniques in **python** and **MATLAB** with a **SQL** codebase
- Participated in the development of **3** different projects, aided by open and constructive discussions with cross-functional teams, engaged in brainstorming sessions to collectively arrive at well-informed decisions

## PROJECTS

**Yank – Search Engine Query Optimizer** May 2025 – July 2025
*AWS, Node.js, Flask, Docker, LLMs, JavaScript, Python, HTML, CSS*

- Architected a browser extension that optimizes search engine results by tackling vague queries to deliver accurate and targeted results, effectively reducing misinterpretations
- Deployed backend services to **AWS EC2** with **AWS Cognito** authentication to handle large-scale concurrent requests with **low p95 latency** to transform queries with additional context based on semantics and measures of vagueness

**MarketMatch – Commercial Matchmaking** April 2025 – June 2025
*Go/Golang, React, Typescript, NoSQL, HTML, CSS*

- Developed a scalable commercial matchmaking platform using **Go** paired with a customized **Gale-Shapley** algorithm to efficiently pair buyers and sellers with data similar to Facebook Marketplace and implemented modern **data security** techniques against **XSS** and **CSRF** attacks
- Engineered a high-performance, real-time user interface with **React** and **TypeScript**, along with **NoSQL** for persistent storage of product listings and user profiles, which enables a seamless swiping feature for product discovery and matching

**SafEE – Personalized Video Player** Jan 2025 – Feb 2025
*HTML, JavaScript, OpenCV, PyAudio, PyQt*

- Conceptualized and engineered a video streaming application for **HTML5** systems that analyzes real-time moving average values of pixel and speaker data to warns users of sudden spikes in audio and video with a small UI integrated dot
- Leveraged **computer vision** and audio output to create a **multi-threaded system** for efficient feedback on a cross-platform frontend framework allowing for access to multiple users

**Deep Learning Applications** November 2023 – December 2023
*PyTorch, TensorFlow*

- Implemented a state-of-the-art **CNN** and **ResNet50** systems to classify lung X-ray images for **COVID-19** and **pneumonic** infections detection among normal cases, achieving an **85%** positive rate measured by **F1 scores**, **Confusion Matrices**, and **ROC** curves
- Utilized **GRU** and **U-net** deep learning architectures to build **translators** and **retina image masking** applications

## SKILLS

**Software Languages:** Python, R, C, C++, Assembly, MATLAB
**Platforms:** Windows, MacOS, Linux
**DevOps:** AWS:EC2, AWS:IAM, Azure:Kubernetes
**Front End:** React, Typescript, JavaScript
**Back End:** Golang, Java, MySQL, AWS:DynamnoDB
**Tools:** Docker, Git, npm